PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR NORTH WESTERN ZONE MOCK EXAMINATION SEP. 2024

031/2B **PHYSICS 2B PRACTICAL**

TIME 2:30 HRS **02/10/2024 WEDNESDAY**

INSTRUCTIONS

01. This paper consist of two (2) question
02. Answer all question
03. Each question carries 25 mark
04. Non- programmable and mathematical table may be allowed
05. Write your examination number on every page of your answer sheet.
06. The following constant may be used
    (i) Gravitational constant g = 10N/kg
    (ii) Pie $A = \pi = 3.14$

01. The aim of this experiment in figure 1 to determine mass Mo of the meter rule provided.

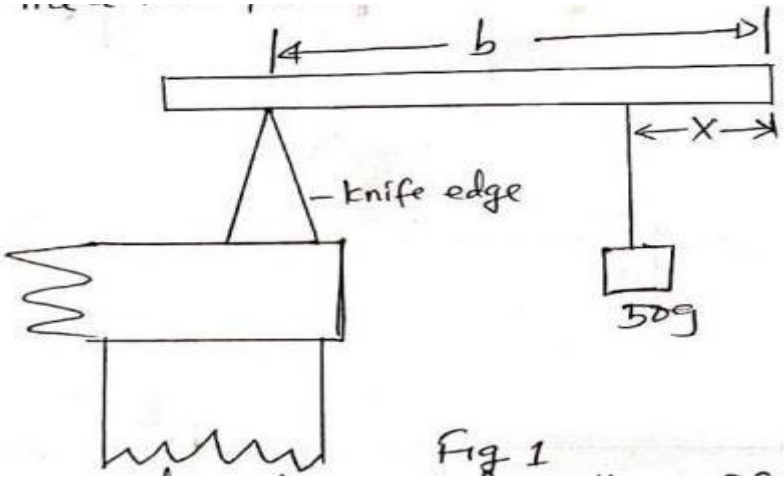


Fig 1

(a) Place a meter rule provided on the knife edge. Balance the meter rule and record the value of Lo

(b) Hang 50g mass to a position where x = 10cm, 15cm, 25cm, 30cm, 40cm, and 45cm, Measure the corresponding value of "b" when the meter rule is balanced.

(c) Plot of the graph of b (on vertical axis) against x (on horizontal axis)

(d) Find the slope of the graph above

(e) From the following relationship $\frac{SMO}{50} + 5 = 1$ where

S = Slope then calculate mass Mo

(f) State any source of error with corresponding precaution to be taken in doing this experiment.

02. Your provided with a resistance box R, a dry cell E, a switch S, a jockey J and a Centre – zero galvanometer. They were required to determine the value of the unknown resistance x by means of a meter bridge as follows

(a) Set up the circuit as shown in figure 2

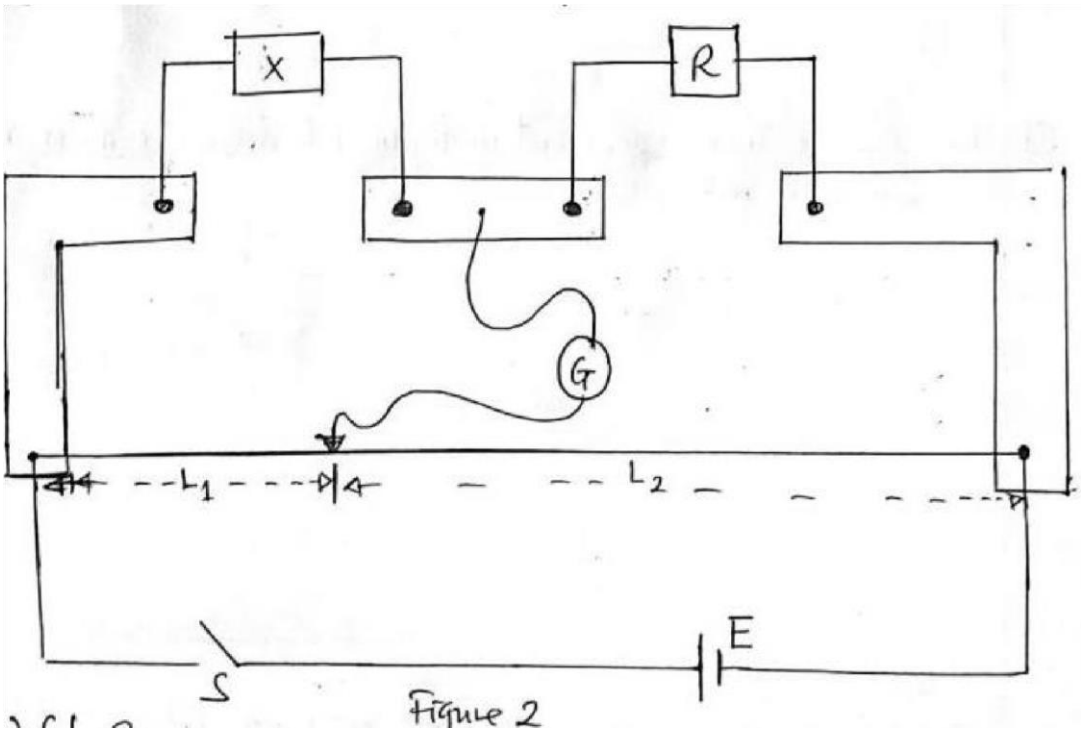


Figure 2

(b) Set R 1Ω, close the switch, slide the jockey over the meter bridge wire until the galvanometer reads zero. Read and record the length L_1. Also read and record the corresponding length L_2

(c) Repeat the procedures in 2 (b) for values of R = 2, 3, 5 and 7 ohms and record the corresponding values for L_1 and L_2 in each case.

QUESTIONS

(i) Tabulate your results including the values of $\frac{L_1}{L_2}$

(ii) Plot a graph of R against $\frac{L_2}{L_1}$

(iii) Find the slope s of the graph

(iv) Determine the value of the unknown resistance by showing clearly how you arrive at your answer.